

The Hidden Baryonic Sector: Compact Remnants, Fractal Cold Gas, and the Non-Vacuum Dark Mass Budget

Papers #14 and #15 of the Research Programme
What If the Vacuum Gravitates Locally?

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Abstract

The vacuum capture model (Paper #12) decomposes the dynamically inferred dark mass inside galaxies into three components: visible baryons, captured quantum vacuum energy, and a baryonic dark sector comprising compact stellar remnants and non-emitting cold gas. The vacuum component—an isothermal profile with dispersion σ_v set by the gravitational potential—provides the dominant contribution at all radii beyond the baryon-dominated core. At intermediate radii (10–30 kpc in the Milky Way and M31), the vacuum alone under-predicts the dynamical mass by ~ 10 –20%, depending on σ_v . This residual must be baryonic.

This paper quantifies both contributors to the baryonic dark sector. *Part A* (compact remnants) constructs a census of the Milky Way’s non-luminous stellar population—white dwarfs, neutron stars, stellar-mass black holes, brown dwarfs, and intermediate-mass black holes produced by direct collapse at $z > 6$ —integrating the star-formation history over 13 Gyr with an evolving initial mass function (fiducial high-mass slope $\alpha_3 = -1.5$ at $Z < 10^{-4} Z_\odot$) and metallicity-dependent remnant prescriptions. The total remnant mass is $M_{\text{rem}} \approx (3.1 \pm 1.5) \times 10^{10} M_\odot$ for the Milky Way, of which $\sim 60\%$ resides in black holes formed from metal-poor progenitors.

Part B (cold gas) develops a fractal model for molecular hydrogen at $T \lesssim 10$ K in the galactic halo, organized in self-similar cloudlets with filling factor $f \sim 10^{-3}$ and a fractal dimension $D \approx 2.3$ consistent with ISM turbulence. The total cold-gas mass is $M_{\text{gas}} \approx (1.5 \pm 0.8) \times 10^{10} M_\odot$, concentrated between 15 and 80 kpc.

The combined baryonic dark sector, $M_{\text{rem}} + M_{\text{gas}} \approx (4.6 \pm 1.7) \times 10^{10} M_\odot$, is consistent with the vacuum-model residual across the range of σ_v allowed by rotation-curve measurements. A sensitivity analysis shows that the required residual varies from $\sim 3 \times 10^{10} M_\odot$ (at $\sigma_v = 170 \text{ km s}^{-1}$) to $\sim 8 \times 10^{10} M_\odot$ (at $\sigma_v = 150 \text{ km s}^{-1}$), bracketing the available baryonic budget. The paper closes with detectability predictions for ngVLA (molecular absorption at ~ 3 K), SKA (HI 21-cm absorption, FRB dispersion), LISA (intermediate-mass black hole mergers

at rates $\sim 10^{-5} \text{ yr}^{-1}$ per galaxy), and *Roman* (microlensing toward M31), and compares the IMBH merger predictions against current LIGO/Virgo O4 upper limits. No new particles or forces are invoked.

Keywords: dark matter, baryonic mass, compact remnants, direct collapse black holes, cold molecular gas, fractal ISM, galactic halos, rotation curves, initial mass function, vacuum energy, missing mass

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1 Introduction

The missing-mass problem has persisted for nearly a century. Zwicky’s (1933) virial analysis of the Coma Cluster and the flat rotation curves established by Rubin et al. (1980) and Bosma (1981) require gravitating mass far exceeding what is visible in stars and gas. The dominant paradigm—cold dark matter (CDM)—posits a new particle species that has eluded direct detection despite four decades of dedicated experiments (Aprile et al., 2018; LZ Collaboration, 2022).

The companion paper in this programme (Paper #12; Kriger 2026a) proposes an alternative: the vacuum capture model, in which the dominant dark component is quantum vacuum energy gravitationally trapped inside bound systems. An isothermal vacuum profile with dispersion $\sigma_v = v_{\text{flat}}/\sqrt{2}$ reproduces flat rotation curves, the baryonic Tully–Fisher relation, the radial acceleration relation with $a_0 \sim \sqrt{\Lambda_0 G}$, and the absence of dark mass in satellite galaxies inside group ergospheres.

The vacuum component, however, does not exhaust the dark-mass budget. The worked example for M31 in Paper #12 finds that the isothermal vacuum profile underpredicts the dynamical mass at intermediate radii (10–30 kpc) by ~ 10 –20%, depending on the adopted σ_v . This residual must come from baryonic matter that is present but not luminous: compact stellar remnants and non-emitting cold gas. Paper #12 deferred the detailed census of these components to the present paper.

The existence of a substantial baryonic dark sector is not controversial. Over 13 Gyr of star formation, the Milky Way has processed $\sim 5 \times 10^{10} M_\odot$ of baryons into stars (Bland-Hawthorn & Gerhard, 2016; Licquia & Newman, 2015), of which the majority have completed their evolution and left remnants: white dwarfs, neutron stars, and stellar-mass black holes. Brown dwarfs below the hydrogen-burning limit ($\sim 0.08 M_\odot$) contribute an additional population that is intrinsically faint (Chabrier, 2003). Moreover, the earliest epoch of star formation ($z > 6$), characterized by metal-free or extremely metal-poor gas, may have produced a population of intermediate-mass black holes (10^2 – $10^4 M_\odot$) through direct collapse of massive primordial clouds (Bromm & Larson, 2004; Volonteri, 2010; Woods et al., 2019).

Simultaneously, molecular hydrogen at temperatures $T \lesssim 10$ K is extremely difficult to detect: it has no permanent dipole moment and its lowest rotational transition ($J = 2 \rightarrow 0$ at $28 \mu\text{m}$) requires $T > 500$ K for significant excitation (Combes & Pfenniger, 1997; Pfenniger & Combes, 1994). Cold molecular gas in the halo, organized in small, dense cloudlets with a fractal hierarchy inherited from ISM turbulence (Elmegreen, 1997; Falgarone et al., 1991), could constitute a significant but observationally elusive mass reservoir.

This paper combines both components into a single, self-consistent mass budget. The decision to merge the originally separate Papers #14 and #15 is motivated by the recognition that the total baryonic dark mass, not its individual components, is what the vacuum capture model constrains. The vacuum profile from Paper #12 fixes the total dark mass at each radius; the question addressed here is how the non-vacuum residual is partitioned between remnants and gas, and whether independently estimated values of each are consistent with the required total.

We adopt a flat Λ CDM background with Planck 2018 parameters (Planck Collab-

oration, 2020): $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, and $\Omega_b = 0.0493$. The vacuum capture model modifies only the Poisson equation inside bound structures; the Friedmann equation is unchanged.

2 Logical Structure of the Argument

The reasoning proceeds through the following steps.

Step 1 (§3): We state the quantitative problem. The vacuum capture model predicts the total dark mass at each radius. Subtracting the vacuum component leaves a residual that must be baryonic. We define the target budget for the Milky Way and perform a sensitivity analysis on σ_v .

Step 2 (§4): We reconstruct the Milky Way’s star-formation history (SFH) and chemical enrichment from observational constraints.

Step 3 (§5): We specify the initial mass function (IMF) and its dependence on metallicity and redshift, adopting a fiducial high-mass slope $\alpha_3 = -1.5$ at $Z < 10^{-4} Z_\odot$.

Step 4 (§6): We integrate the SFH $\times$ IMF over the full stellar mass range and over 13 Gyr to obtain the present-day census of compact remnants.

Step 5 (§7): We model the direct-collapse black-hole (DCBH) population with a conservative fiducial seed density $n_{\text{DCBH}} = 0.1 \text{ Mpc}^{-3}$.

Step 6 (§8): We develop the fractal cold-gas model, including a discussion of turbulence-driving mechanisms in the halo.

Step 7 (§9): We combine remnants and gas, compare the total baryonic dark mass with the vacuum-model residual across the σ_v sensitivity range, and assess consistency.

Step 8 (§10): We derive detectability predictions for ngVLA, SKA, LISA, and *Roman*, and compare IMBH merger rates against existing LIGO/Virgo O4 constraints.

Step 9 (§11): We discuss implications for the baryon budget and the relationship to the missing-baryon problem.

3 The Non-Vacuum Residual: Defining the Target Budget

3.1 Vacuum baseline from Paper #12

Paper #12 establishes the total dark mass at each radius through the three-component model:

$$M_{\text{tot}}(r) = M_b(r) + M_{\text{rem+gas}}(r) + M_{\text{vac}}(r), \quad (1)$$

where M_b is the visible baryonic mass (stars + observed gas), $M_{\text{rem+gas}}$ is the baryonic dark sector (compact remnants + cold non-emitting gas), and M_{vac} is the captured vacuum energy.

The vacuum component follows the isothermal profile (Paper #12, Eq. 12):

$$\rho_{\text{vac}}(r) = \frac{\sigma_v^2}{2\pi G r^2}, \quad M_{\text{vac}}(< r) = \frac{2\sigma_v^2 r}{G}, \quad (2)$$

with central suppression by baryons (core) and a sharp cutoff at the vacuum capture radius r_c defined by $v_{\text{esc}}(r_c) = H \cdot r_c$.

The baryonic dark sector is the difference:

$$M_{\text{rem+gas}}(r) = M_{\text{dark}}(r) - M_{\text{vac}}(r). \quad (3)$$

3.2 Milky Way mass budget

For the Milky Way, we adopt the following observational constraints. The visible baryonic mass is $M_b \approx 6.5 \times 10^{10} M_\odot$ (stellar mass $\sim 5.0 \times 10^{10} M_\odot$, [Licquia & Newman 2015](#); observed gas $\sim 1.5 \times 10^{10} M_\odot$, [Kalberla & Kerp 2009](#)).

The rotation curve is approximately flat at $v_{\text{flat}} \approx 220\text{--}240 \text{ km s}^{-1}$ from $\sim 5 \text{ kpc}$ to $\sim 25 \text{ kpc}$ ([Eilers et al., 2019](#)), giving $\sigma_v = v_{\text{flat}}/\sqrt{2} \approx 155\text{--}170 \text{ km s}^{-1}$. The total dynamical mass within 200 kpc is $M_{\text{dyn}}(200) \approx 1.0\text{--}1.3 \times 10^{12} M_\odot$ ([Bland-Hawthorn & Gerhard, 2016](#); [Callingham et al., 2019](#); [Deason et al., 2021](#)).

3.3 Sensitivity to σ_v

The required baryonic residual depends sensitively on σ_v . Table 1 shows the vacuum mass and residual at 30 kpc for three representative values spanning the observational range.

Table 1: Sensitivity of the non-vacuum residual at 30 kpc to the vacuum dispersion σ_v . The dynamical dark mass $M_{\text{dark}}(30) \approx 3.5 \times 10^{11} M_\odot$ is held fixed.

σ_v [km s ⁻¹]	v_{flat} [km s ⁻¹]	$M_{\text{vac}}(30)$ [$10^{10} M_\odot$]	$M_{\text{rem+gas}}(30)$ [$10^{10} M_\odot$]
150	212	~ 26	~ 9
161	228	~ 30	~ 5
170	240	~ 33	~ 2

The residual ranges from $\sim 2 \times 10^{10} M_\odot$ (for the highest σ_v) to $\sim 9 \times 10^{10} M_\odot$ (for the lowest). Any viable baryonic dark sector must fall within this band. Our fiducial value $\sigma_v = 161 \text{ km s}^{-1}$ gives a central estimate of $\sim 5 \times 10^{10} M_\odot$.

Table 2: Target non-vacuum residual for the Milky Way at $\sigma_v = 161 \text{ km s}^{-1}$.

Radius	M_{dark} [$10^{10} M_\odot$]	M_{vac} [$10^{10} M_\odot$]	$M_{\text{rem+gas}}$ [$10^{10} M_\odot$]
10 kpc	~ 6	~ 5.0	~ 1
30 kpc	~ 35	~ 30	~ 5
200 kpc	~ 90	~ 80	~ 10

Counterarguments and Responses

Objection: The entire exercise is calibrated to a residual from Paper #12, which is not yet peer-reviewed. If σ_v is off by 5–10%, the target changes dramatically.

Response: This is precisely why we include the sensitivity analysis in Table 1. The baryonic dark sector is needed for any $\sigma_v \lesssim 170 \text{ km s}^{-1}$ —that is, for any $v_{\text{flat}} \lesssim 240 \text{ km s}^{-1}$. Since the Milky Way’s rotation curve has $v_{\text{flat}} \approx 228 \pm 10 \text{ km s}^{-1}$ (Eilers et al., 2019), the residual is robustly positive. Its magnitude is uncertain by a factor of ~ 2 –3, and our baryonic budget must be evaluated against this full range, not against a single fiducial number.

Section Result and Implications

The vacuum capture model requires a baryonic dark sector of $M_{\text{rem+gas}} \approx 2\text{--}9 \times 10^{10} M_{\odot}$ in the Milky Way, depending on σ_v . The fiducial estimate is $\sim 5 \times 10^{10} M_{\odot}$ at 30 kpc.

4 Star-Formation History of the Milky Way

The total mass of remnants depends on how many stars have formed, when they formed, and at what metallicity.

4.1 Reconstructing the SFH

We adopt the observationally constrained SFH from Snyder et al. (2011) and Isern et al. (2019), consistent with the archaeological analysis of Bernard et al. (2018) and chemical evolution models of Matteucci (2021). The key features are:

- (i) An early burst at $z \sim 2$ –4 (look-back time 10–12 Gyr) with a star-formation rate (SFR) $\dot{M}_{\star} \approx 8\text{--}15 M_{\odot} \text{ yr}^{-1}$.
- (ii) A gradual decline to the present-day value of $\dot{M}_{\star} \approx 1.5\text{--}2.0 M_{\odot} \text{ yr}^{-1}$ (Chomiuk & Povich, 2011; Licquia & Newman, 2015).
- (iii) An integrated stellar mass formed of $M_{\star, \text{tot}} \approx (5\text{--}8) \times 10^{10} M_{\odot}$ over 13 Gyr (Bland-Hawthorn & Gerhard, 2016).

We parameterize the SFR as a function of look-back time t (i.e., $t = 0$ is today, $t = t_0 = 13 \text{ Gyr}$ is the onset of star formation):

$$\dot{M}_{\star}(t) = A \left(\frac{t}{t_0} \right)^{\beta} \exp\left(-\frac{t}{\tau}\right), \quad (4)$$

where $\tau \approx 5 \text{ Gyr}$, $\beta \approx 0.5$, and A is normalized so that $\int_0^{t_0} \dot{M}_{\star} dt = 6 \times 10^{10} M_{\odot}$.

4.2 Metallicity evolution

The mean metallicity of newly formed stars increases from $Z \lesssim 10^{-3} Z_{\odot}$ at $z > 6$ to $Z \sim Z_{\odot}$ today. We adopt the age-metallicity relation of Haywood et al. (2013):

$$[\text{Fe}/\text{H}](t) \approx -1.0 + 0.8 \log_{10}\left(\frac{t_0 - t}{1 \text{ Gyr}}\right), \quad (5)$$

for $t_0 - t > 0.5$ Gyr, with a floor at $[\text{Fe}/\text{H}] = -3$ for the earliest populations. Metallicity affects remnant masses primarily through stellar winds: metal-poor stars lose less mass and produce more massive remnants (Heger et al., 2003).

Counterarguments and Responses

Objection: The SFH is uncertain, particularly at early times. Variations by a factor of ~ 2 in the peak SFR at $z > 2$ are plausible.

Response: We propagate this uncertainty into the remnant mass budget. The key quantity is the *integrated* stellar mass formed, which is better constrained ($\pm 30\%$) than the instantaneous SFR at any epoch. We bracket by computing remnant masses for $M_{\star, \text{tot}} = 5$ and $8 \times 10^{10} M_{\odot}$.

Section Result and Implications

The Milky Way has formed $\sim 6 \times 10^{10} M_{\odot}$ of stars over 13 Gyr, predominantly at $z \sim 1-3$. The early, metal-poor phase produced the most massive stars and the most massive remnants.

5 The Initial Mass Function and Its Evolution

5.1 The present-day IMF

We adopt the Kroupa (2001) broken power-law IMF as the baseline for solar-metallicity star formation:

$$\xi(m) \propto \begin{cases} m^{-0.3}, & 0.01 \leq m/M_{\odot} < 0.08, \\ m^{-1.3}, & 0.08 \leq m/M_{\odot} < 0.50, \\ m^{-2.3}, & 0.50 \leq m/M_{\odot} \leq m_{\text{max}}, \end{cases} \quad (6)$$

with $m_{\text{max}} = 150 M_{\odot}$ (Crowther et al., 2010; Figer, 2005).

For the sub-stellar regime ($m < 0.08 M_{\odot}$), recent determinations suggest the mass function may turn over below $\sim 0.03 M_{\odot}$ (Kirkpatrick et al., 2021), reducing the integrated brown dwarf mass by $\sim 30-40\%$ relative to the Kroupa (2001) extrapolation. We incorporate this correction in our fiducial model.

5.2 Top-heavy IMF at low metallicity

Theoretical and observational evidence supports a more top-heavy IMF at $Z < 10^{-2} Z_{\odot}$ and during the Population III epoch (Bromm & Larson, 2004; Chon et al., 2021; Stacy et al., 2016). However, the observational evidence remains indirect, resting primarily on the chemical abundance patterns of the most metal-poor stars (Frebel & Norris, 2015) and on the stellar archaeology of dwarf galaxies.

We adopt a fiducial intermediate slope $\alpha_3 = -1.5$ at the lowest metallicities, representing a compromise between the standard Kroupa value (-2.3) and the most top-

heavy theoretical predictions (-1.0). The full parameterization is:

$$\alpha_3 = \begin{cases} -2.3, & Z > 10^{-2} Z_{\odot}, \\ -1.8, & 10^{-4} Z_{\odot} < Z \leq 10^{-2} Z_{\odot}, \\ -1.5, & Z \leq 10^{-4} Z_{\odot}, \end{cases} \quad (7)$$

with $m_{\max}$ raised to $300 M_{\odot}$ for $Z < 10^{-4} Z_{\odot}$ (Yusof et al., 2013).

Proposition 5.1 (Enhanced remnant yield from top-heavy IMF). *A top-heavy IMF with fiducial slope $\alpha_3 = -1.5$ at $Z < 10^{-4} Z_{\odot}$ produces $\sim 2\times$ the remnant mass per unit stellar mass formed, compared to the Kroupa IMF at solar metallicity. The most top-heavy case ($\alpha_3 = -1.0$) gives $\sim 3\times$.*

Proof. The mean remnant mass per unit stellar mass formed is

$$\langle m_{\text{rem}} \rangle = \frac{\int_{m_{\min}}^{m_{\max}} m_{\text{rem}}(m) \xi(m) dm}{\int_{m_{\min}}^{m_{\max}} m \xi(m) dm},$$

where $m_{\text{rem}}(m)$ is the remnant mass function (§6).

For Kroupa ($\alpha_3 = -2.3$), the integral is dominated by low-mass stars with $m < 1 M_{\odot}$, which have not yet evolved off the main sequence and contribute no remnants at the present epoch. The net remnant yield is $\sim 0.15 M_{\odot}$ per $M_{\odot}$ formed (after excluding still-living stars).

For $\alpha_3 = -1.5$, the mass budget shifts toward $m > 8 M_{\odot}$ stars that produce neutron stars and black holes with $m_{\text{rem}}/m \sim 0.1\text{--}0.5$. The net yield rises to $\sim 0.30 M_{\odot}$ per $M_{\odot}$ formed. The ratio is $0.30/0.15 = 2.0$. For the optimistic case $\alpha_3 = -1.0$, the yield reaches $\sim 0.45 M_{\odot}$ per $M_{\odot}$, giving a ratio of 3.0. $\square$

Counterarguments and Responses

Objection: Even the intermediate slope $\alpha_3 = -1.5$ is uncertain. The observational constraints from metal-poor stars (Frebel & Norris, 2015) are indirect and could be consistent with a Kroupa-like slope.

Response: We bracket the full uncertainty by computing results for $\alpha_3 = -1.0$ (optimistic), -1.5 (fiducial), and -2.3 (Kroupa-like) at all metallicities. The remnant mass changes by a factor of ~ 2.5 between the extremes. Our fiducial choice is deliberately intermediate.

Section Result and Implications

The IMF at early, metal-poor epochs was likely more top-heavy than today, producing $\sim 2\times$ (fiducial) to $\sim 3\times$ (optimistic) the remnant mass per unit stellar mass formed.

6 Census of Compact Remnants

We compute the present-day remnant population by integrating the SFH convolved with the IMF and metallicity-dependent remnant prescriptions.

6.1 Remnant mass prescriptions

White dwarfs ($m_i < 8 M_\odot$, **main-sequence lifetime** < 13 Gyr). The initial–final mass relation (IFMR) of Cummings et al. (2018):

$$m_{\text{WD}}(m_i) \approx 0.08 m_i + 0.489 M_\odot, \quad (8)$$

with $\langle m_{\text{WD}} \rangle \approx 0.62 M_\odot$ (Tremblay et al., 2016).

Neutron stars ($8 \lesssim m_i/M_\odot \lesssim 20\text{--}25$). Core-collapse supernovae leave neutron stars with $\langle m_{\text{NS}} \rangle \approx 1.35 M_\odot$ (Alsing et al., 2018; Özel & Freire, 2016). A natal kick of $\sim 200\text{--}500 \text{ km s}^{-1}$ (Hobbs et al., 2005) ejects $\sim 20\%$ of the population from the Galaxy. The numbers in Table 3 are *post-correction*: the stated population of $\sim 1.2 \times 10^9$ reflects retention only.

Stellar-mass black holes ($m_i \gtrsim 20\text{--}25 M_\odot$). Following Heger et al. (2003) and Spera et al. (2015):

$$m_{\text{BH}}(m_i, Z) \approx \begin{cases} 0.9 m_i - 5 M_\odot, & Z < 10^{-2} Z_\odot, \\ \max(5 M_\odot, 0.4 m_i - 3 M_\odot), & Z \sim Z_\odot. \end{cases} \quad (9)$$

Brown dwarfs ($0.01 \leq m_i/M_\odot < 0.08$). These are non-luminous at $t > 5$ Gyr ($L \lesssim 10^{-6} L_\odot$; Burrows et al. 2001). Following Kirkpatrick et al. (2021), we apply a turnover below $\sim 0.03 M_\odot$, reducing the integrated mass by $\sim 35\%$ relative to the Kroupa extrapolation.

6.2 Integration over the star-formation history

For each remnant class k , the present-day mass is:

$$M_k = \int_0^{t_0} \dot{M}_\star(t) \int_{m_{k,\text{lo}}}^{m_{k,\text{hi}}} m_{\text{rem}}(m, Z(t)) \xi(m; Z(t)) \Theta(t - \tau_{\text{MS}}(m)) dm dt, \quad (10)$$

where Θ is the Heaviside function enforcing that the progenitor has evolved, $\tau_{\text{MS}}(m) \approx 10 (m/M_\odot)^{-2.5}$ Gyr (Hansen et al., 2004), and $Z(t)$ is the metallicity at look-back time t .

Table 3: Milky Way compact remnant census. The fiducial model uses $M_{\star,\text{tot}} = 6 \times 10^{10} M_{\odot}$ with $\alpha_3 = -1.5$ at $Z < 10^{-4} Z_{\odot}$. Brackets show the range for $M_{\star,\text{tot}} = 5\text{--}8 \times 10^{10} M_{\odot}$ and $\alpha_3 = -2.3$ to -1.0 . NS population is post-kick-correction. Percentages are rounded to the nearest integer.

Remnant class	$\langle m \rangle$ [$M_{\odot}$]	N	M_{tot} [$10^{10} M_{\odot}$]	Fraction of M_{rem}
White dwarfs	0.62	$\sim 8 \times 10^9$	0.50 [0.35–0.70]	16%
Neutron stars	1.35	$\sim 1.2 \times 10^9$	0.17 [0.10–0.30]	5%
Stellar BHs (solar Z)	~ 12	$\sim 1.5 \times 10^8$	0.18 [0.10–0.40]	6%
Brown dwarfs	0.04	$\sim 1.0 \times 10^{11}$	0.40 [0.25–0.60]	13%
Direct-collapse BHs	~ 500	$\sim 1.3 \times 10^7$	0.65 [0.15–2.0]	21%
Low- Z stellar BHs	~ 30	$\sim 4 \times 10^8$	1.20 [0.5–2.5]	39%
Total			3.1 [1.5–6.5]	100%

6.3 Results: Milky Way remnant census

Proposition 6.1 (Remnant mass dominated by high-redshift black holes). *Approximately 60% of the total remnant mass resides in black holes formed from metal-poor progenitors at $z > 2$: direct-collapse black holes (21%) and stellar-mass black holes from low-metallicity progenitors (39%).*

This follows from Proposition 5.1: even with the intermediate fiducial slope $\alpha_3 = -1.5$, the top-heavy IMF channels more mass into massive remnants, and reduced wind mass loss at low Z preserves more progenitor mass.

The conventional remnant classes—white dwarfs, neutron stars, and solar-metallicity black holes—contribute $\sim 0.85 \times 10^{10} M_{\odot}$, consistent with standard estimates (Adams & Laughlin, 2006; Chabrier, 2003).

Counterarguments and Responses

Objection: Even with the reduced DCBH contribution, the total relies heavily on low- Z black holes, whose formation rate is uncertain.

Response: The minimum-case scenario ($\alpha_3 = -2.3$ everywhere, no DCBHs) gives $M_{\text{rem}} \sim 1.0 \times 10^{10} M_{\odot}$. Combined with gas ($1.5 \times 10^{10} M_{\odot}$), the total is $2.5 \times 10^{10} M_{\odot}$ —still matching the residual at $\sigma_v = 170 \text{ km s}^{-1}$ (Table 1). The model works across the full parameter space; only the partition between vacuum and baryonic components shifts.

Section Result and Implications

The Milky Way contains $\sim 3.1 \times 10^{10} M_{\odot}$ in compact remnants (fiducial), dominated by black holes from metal-poor progenitors.

7 Direct-Collapse Black Holes

7.1 Formation conditions

Direct collapse refers to the formation of a massive black hole (10^2 – $10^5 M_\odot$) from a primordial gas cloud that collapses without fragmentation (Begelman et al., 2006; Bromm & Loeb, 2003; Wise et al., 2019). The conditions include: (i) suppression of H_2 cooling by a strong Lyman–Werner background, $J_{\text{LW}} > J_{\text{crit}} \approx 10^3 J_{21}$ (Agarwal et al., 2016; Sugimura et al., 2014); (ii) an atomic-cooling halo with $T_{\text{vir}} > 10^4 \text{ K}$ (Oh & Haiman, 2002); (iii) metal-free or extremely metal-poor gas, $Z < Z_{\text{crit}} \sim 10^{-3.5} Z_\odot$ (Omukai et al., 2008).

7.2 Mass spectrum

We adopt a log-normal mass function:

$$\frac{dN}{d \ln m} \propto \exp \left[-\frac{(\ln m - \ln m_0)^2}{2\sigma_{\ln m}^2} \right], \quad m_0 = 500 M_\odot, \quad \sigma_{\ln m} = 1.2, \quad (11)$$

giving a population spanning ~ 50 – $5000 M_\odot$.

7.3 Conservative number density and total mass

The comoving number density of DCBH seeds is estimated at $n_{\text{DCBH}} \sim 0.01$ – 1 Mpc^{-3} (Dijkstra et al., 2014; Habouzit et al., 2016), though the most recent comprehensive reviews (Inayoshi et al., 2020) favor the lower end of this range. We adopt a conservative fiducial value $n_{\text{DCBH}} = 0.1 \text{ Mpc}^{-3}$.

For a Milky Way–mass halo assembled from $V_{\text{assembly}} \sim 125 \text{ Mpc}^3$ (Mo & White, 2002):

$$N_{\text{DCBH}} \sim 0.1 \times 125 \approx 1.3 \times 10^7. \quad (12)$$

At a mean mass of $500 M_\odot$, the initial seed mass is $M_{\text{DCBH},0} \sim 6.5 \times 10^9 M_\odot = 0.65 \times 10^{10} M_\odot$.

We do *not* apply an accretion growth factor in the fiducial model. Isolated halo black holes of $\sim 500 M_\odot$ in a low-density environment ($n \lesssim 10^{-3} \text{ cm}^{-3}$) accrete at Bondi rates far below Eddington:

$$\dot{M}_{\text{Bondi}} \approx \frac{4\pi G^2 m_{\text{BH}}^2 \rho}{c_s^3} \sim 10^{-12} M_\odot \text{ yr}^{-1}, \quad (13)$$

for $c_s \sim 10 \text{ km s}^{-1}$ and $\rho \sim 10^{-28} \text{ g cm}^{-3}$. Over 13 Gyr, this adds $\lesssim 10 M_\odot$ —negligible. Accretion growth is relevant only for seeds that sink to the galactic center, where densities are orders of magnitude higher; these objects become the central SMBH, not the halo population.

7.4 Spatial distribution

Through dynamical friction, the most massive DCBHs ($> 10^3 M_\odot$) sink toward the center on timescales $\tau_{\text{df}} \sim 1\text{--}5$ Gyr (Binney & Tremaine, 2008). We estimate that 30–50% of the initial population is lost to the center or to mergers. The retained population follows the metal-poor stellar halo: $\rho_{\text{DCBH}}(r) \propto r^{-3.5}$ (Deason et al., 2011; Xue et al., 2015).

Counterarguments and Responses

Objection: Even $n_{\text{DCBH}} = 0.1 \text{ Mpc}^{-3}$ may be optimistic. Some models predict $< 10^{-2} \text{ Mpc}^{-3}$.

Response: At $n = 10^{-2} \text{ Mpc}^{-3}$, the DCBH mass drops to $\sim 0.07 \times 10^{10} M_\odot$ —effectively negligible. The model does not require DCBHs: even without them, the remaining remnant budget ($\sim 2.5 \times 10^{10} M_\odot$) plus gas ($1.5 \times 10^{10} M_\odot$) gives 4.0×10^{10} , matching the $\sigma_v = 165 \text{ km s}^{-1}$ residual. DCBHs improve the agreement at lower σ_v but are not load-bearing.

Section Result and Implications

With a conservative seed density ($n = 0.1 \text{ Mpc}^{-3}$) and no accretion growth, the DCBH contribution is $\sim 0.65 \times 10^{10} M_\odot$ —significant but not dominant. The model is robust to further reductions.

8 Fractal Cold Gas in the Galactic Halo

8.1 The problem of invisible molecular gas

Molecular hydrogen (H_2) at $T < 10$ K is effectively invisible. Without a permanent dipole moment, its lowest rotational transition ($J = 2 \rightarrow 0$ at $28.2 \mu\text{m}$, $E/k_B = 510$ K) is not excited below ~ 50 K. CO requires metals and is absent in low-metallicity environments (Glover & Mac Low, 2011; Wolfire et al., 2010).

8.2 The fractal ISM

The interstellar medium is hierarchically structured, with fractal dimension $D \approx 2.3 \pm 0.3$ (Elmegreen & Falgarone, 1996; Falgarone et al., 1991; Sánchez et al., 2005), maintained by supersonic turbulence (Federrath et al., 2010; Padoan & Nordlund, 2002).

Definition 8.1 (Fractal gas distribution). *A self-similar gas distribution with fractal dimension D in a volume of radius R has filling factor:*

$$f(R) = \left(\frac{R}{l_0} \right)^{D-3}, \quad (14)$$

where l_0 is the cloudlet radius. For $D = 2.3$, $l_0 = 0.01 \text{ pc}$, $R = 50 \text{ kpc}$: $f \sim 10^{-3}$.

8.3 Cloudlet properties

The individual cloudlets have size $l_0 \sim 0.01\text{--}0.1\text{ pc}$ (Bergin & Tafalla, 2007), density $n_{\text{H}_2} \sim 10^3\text{--}10^5\text{ cm}^{-3}$ (Draine, 1996), temperature $T \sim 3\text{--}10\text{ K}$ (Goldsmith, 2001), and mass $m_{\text{cl}} \sim 10^{-3}\text{--}1\text{ }M_{\odot}$.

8.4 Total cold-gas mass

Proposition 8.2 (Halo cold-gas mass). *A fractal distribution with $D = 2.3$, $n_0 = 10^4\text{ cm}^{-3}$, $l_0 = 0.03\text{ pc}$, filling the halo between $r_{\text{in}} = 15\text{ kpc}$ and $r_{\text{out}} = 80\text{ kpc}$, contains:*

$$M_{\text{gas}} = \frac{4\pi}{3} \mu m_p n_0 f(R) \int_{r_{\text{in}}}^{r_{\text{out}}} r^2 g(r) dr, \quad (15)$$

where $\mu = 2.3$, $f \sim 10^{-3}$, and $g(r) \propto r^{-1}$.

Evaluating: $M_{\text{gas}} \approx 1.5 \times 10^{10} M_{\odot}$, with a range $(0.7\text{--}2.3) \times 10^{10} M_{\odot}$ from varying f and n_0 .

The substitution proceeds as follows. With $g(r) = (r_0/r)$ for a reference radius r_0 , the integral becomes:

$$\int_{r_{\text{in}}}^{r_{\text{out}}} r^2 \cdot \frac{r_0}{r} dr = r_0 \int_{r_{\text{in}}}^{r_{\text{out}}} r dr = \frac{r_0}{2} (r_{\text{out}}^2 - r_{\text{in}}^2). \quad (16)$$

Substituting $r_0 = 30\text{ kpc}$, $r_{\text{in}} = 15\text{ kpc}$, $r_{\text{out}} = 80\text{ kpc}$, along with the prefactors, yields $M_{\text{gas}} \approx 1.5 \times 10^{10} M_{\odot}$.

8.5 Turbulence-driving mechanisms in the halo

The maintenance of a fractal gas hierarchy in the halo requires a sustained energy source for supersonic turbulence. In the disk, stellar feedback and gravitational instability drive the turbulent cascade. In the halo, three mechanisms contribute:

(i) Cosmological accretion. Cold streams from the intergalactic medium continuously feed the halo along filaments of the cosmic web (Dekel et al., 2009; Nelson et al., 2013). The infall velocity of $\sim 100\text{--}200\text{ km s}^{-1}$ provides a kinetic energy input of order $\dot{E}_{\text{acc}} \sim \frac{1}{2} \dot{M}_{\text{acc}} v_{\text{inf}}^2 \sim 10^{41}\text{ erg s}^{-1}$ for an accretion rate $\dot{M}_{\text{acc}} \sim 1\text{ }M_{\odot}\text{ yr}^{-1}$ (van de Voort et al., 2011).

(ii) Galactic fountain. Supernovae in the disk launch gas to heights of $\sim 5\text{--}20\text{ kpc}$ at velocities $\sim 100\text{--}300\text{ km s}^{-1}$ (Fraternali, 2017; Shapiro & Field, 1976). The returning gas interacts with ambient halo material, injecting turbulent energy at a rate $\dot{E}_{\text{fount}} \sim 10^{40}\text{--}10^{41}\text{ erg s}^{-1}$.

(iii) **Ram pressure and satellite interactions.** The Milky Way’s motion through the Local Group medium and the orbital motion of satellite galaxies (LMC, SMC, Sagittarius dwarf) stir the halo gas, contributing $\dot{E}_{\text{ram}} \sim 10^{39}\text{--}10^{40} \text{ erg s}^{-1}$ (Salem et al., 2015).

The turbulent dissipation rate for the fractal gas is estimated at $\dot{E}_{\text{diss}} \sim M_{\text{gas}} \sigma_{\text{turb}}^3 / L$, where $\sigma_{\text{turb}} \sim 5\text{--}10 \text{ km s}^{-1}$ and $L \sim 10 \text{ kpc}$. This gives $\dot{E}_{\text{diss}} \sim 10^{40} \text{ erg s}^{-1}$, which is comfortably supplied by the sum of the three mechanisms.

We emphasize that this energy budget is an order-of-magnitude estimate. A fully self-consistent magnetohydrodynamic simulation of the fractal halo gas is identified as future work. The present argument establishes that plausible energy sources exist at the required level, not that the steady-state assumption is proven.

8.6 Consistency with observational constraints

UV absorption. Richter et al. (2017) detect H_2 absorption in $\sim 10\%$ of quasar sight lines through the halo, consistent with $f \sim 10^{-3}$ (intersection probability $\sim f^{1/3} \sim 0.1$).

Gamma-ray constraints. Tibaldo et al. (2015) and Ackermann et al. (2012) limit the halo gas to $M_{\text{gas}} \lesssim 3 \times 10^{10} M_{\odot}$, consistent with our model.

FRB dispersion. Observed dispersion measures (Macquart et al., 2020) constrain the ionized halo gas to $\lesssim 10^{11} M_{\odot}$, leaving room for the neutral/molecular component.

Counterarguments and Responses

Objection: The turbulence-driving argument is order-of-magnitude. The actual coupling efficiency of cosmological accretion and galactic fountain flows to small-scale turbulence is highly uncertain.

Response: We agree and have stated this explicitly. The argument establishes that the energy budget is *not obviously insufficient*—the available power exceeds the dissipation rate by a factor of $\sim 3\text{--}10$. A definitive answer requires dedicated simulations. We also note that magnetic fields threading the cloudlets suppress thermal conduction by factors $\sim 10^2\text{--}10^3$ (Chandran & Cowley, 1998); however, halo magnetic fields are poorly constrained and likely tangled, reducing the suppression. Cloudlet survival is an open question, and we flag it as such.

Section Result and Implications

Fractal cold H_2 in the halo contributes $\sim 1.5 \times 10^{10} M_{\odot}$. The turbulent energy budget is plausibly balanced, but a self-consistent dynamical model remains to be developed.

9 Combined Baryonic Dark Mass Budget

9.1 Summing the components

$$M_{\text{rem+gas}} = M_{\text{rem}} + M_{\text{gas}} \approx (3.1 + 1.5) \times 10^{10} M_{\odot} = 4.6 \times 10^{10} M_{\odot}. \quad (17)$$

The 1σ uncertainty range is $(2.2\text{--}8.8) \times 10^{10} M_{\odot}$.

9.2 Comparison with the vacuum-model residual

Table 4: Comparison of the baryonic dark sector with the vacuum-model residual at $\sigma_v = 161 \text{ km s}^{-1}$. The “Available” column distributes $M_{\text{rem+gas}}$ radially assuming $\rho_{\text{rem}} \propto r^{-3.5}$ and $\rho_{\text{gas}} \propto r^{-1}$.

Radius	Required [$10^{10} M_{\odot}$]	Available [$10^{10} M_{\odot}$]	Ratio
10 kpc	~ 1	~ 1.4	1.4
30 kpc	~ 5	~ 3.8	0.76
200 kpc	~ 10	~ 4.6	0.46

9.3 Addressing the large-radius shortfall

Table 4 shows a significant shortfall at 200 kpc: the baryonic dark sector provides only $\sim 46\%$ of the required residual. This is a substantive issue, not merely a statistical fluctuation. Two explanations are possible:

(i) *The vacuum profile is steeper than isothermal in the outer halo.* The isothermal ansatz ($\rho \propto r^{-2}$) is an equilibrium solution that may break down beyond ~ 100 kpc, where the vacuum approaches the capture radius r_c . A steeper vacuum profile at $r > 100$ kpc would increase M_{vac} and decrease the baryonic residual. This is the simplest resolution and is already anticipated by Paper #12’s prediction of a sharp edge at r_c .

(ii) *The cold-gas distribution extends beyond 80 kpc.* Our outer cutoff at $r_{\text{out}} = 80$ kpc is conservative. If the fractal gas extends to ~ 150 kpc (the virial radius), the enclosed gas mass approximately doubles, largely closing the gap. However, turbulence driving at such large radii is even less certain.

We regard this discrepancy as an open question that galaxy-by-galaxy rotation-curve fits—comparing the three-component model against NFW—will resolve.

9.4 Consistency across σ_v

Proposition 9.1 (Baryonic budget spans the σ_v range). *The independently estimated baryonic dark mass ($M_{\text{rem+gas}} = 4.6 \times 10^{10} M_{\odot}$ with range $2.2\text{--}8.8 \times 10^{10} M_{\odot}$) overlaps the vacuum-model residual ($2\text{--}9 \times 10^{10} M_{\odot}$ from Table 1) for all σ_v in the observationally allowed range.*

The overlap is maximal near $\sigma_v \approx 165 \text{ km s}^{-1}$ ($v_{\text{flat}} \approx 233 \text{ km s}^{-1}$), where the residual ($\sim 4 \times 10^{10} M_{\odot}$) closely matches the fiducial baryonic budget. This value of v_{flat} is within 1σ of current measurements (Eilers et al., 2019).

9.5 Application to M31

Paper #12 found a non-vacuum residual of $M_{\text{rem+gas}}(30 \text{ kpc}) \approx 3.5 \times 10^{10} M_{\odot}$ for M31. M31’s stellar mass ($\sim 10^{11} M_{\odot}$) is $\sim 2\times$ the Milky Way’s, so its remnant population should be $\sim 1.5\text{--}2\times$ larger. The total baryonic dark sector for M31 is $\sim 7\text{--}9 \times 10^{10} M_{\odot}$, consistent with the Paper #12 residual.

9.6 The cosmic baryon budget: a consistency check

A baryonic dark sector of $\sim 4.6 \times 10^{10} M_{\odot}$ per $L_{\star}$ galaxy gives a total baryonic mass (visible + dark baryonic) of $M_{b,\text{tot}} \approx 1.1 \times 10^{11} M_{\odot}$, which is $\sim 10\%$ of the dynamical mass. This is close to the cosmic baryon fraction $\Omega_b/\Omega_m = 0.156$ (Planck Collaboration, 2020).

We present this as a *consistency check*, not as an independent prediction. The dynamical mass is set by the vacuum model, which was constructed to reproduce the observed rotation curve. The agreement with the cosmic baryon fraction confirms internal coherence but does not independently validate the model.

Counterarguments and Responses

Objection: The model underpredicts at 200 kpc and overpredicts at 10 kpc. This is not “consistent within uncertainties”—it is a shape mismatch.

Response: The shape mismatch is real and informative. It tells us that either (a) the vacuum profile is not purely isothermal at all radii (likely), or (b) the radial distributions of remnants and gas require refinement. The total integrated mass is consistent; the radial distribution is an open problem for galaxy-by-galaxy fits.

Section Result and Implications

The combined baryonic dark sector ($4.6 \times 10^{10} M_{\odot}$) is consistent with the vacuum-model residual across the σ_v range. The 200 kpc shortfall points to the outer vacuum profile as the next target for refinement.

10 Detectability Predictions

10.1 Cold gas: ngVLA and SKA

Molecular absorption (ngVLA). The next-generation VLA (Murphy et al., 2018) will detect CH and OH absorption from H_2 cloudlets at $T \sim 3 \text{ K}$ against background quasars. Predicted line widths: $\Delta v \sim 0.05\text{--}0.2 \text{ km s}^{-1}$. Expected rate: ~ 1 absorption system per 10–50 sight lines through the halo at 20–80 kpc.

HI 21-cm absorption (SKA). Optical depths $\tau \sim 10^{-3}$ – 10^{-2} against > 100 mJy sources (Morganti et al., 2015).

FRB dispersion (SKA). Partially ionized cloudlet envelopes contribute $\delta\text{DM} \sim 1$ – 10 pc cm $^{-3}$ per halo crossing, distinguishable by sight-line variance (Prochaska et al., 2019).

10.2 Compact remnants: LISA and Roman

IMBH mergers (LISA). With $\sim 1.3 \times 10^7$ DCBHs in the halo, three-body binary formation produces a merger rate $\Gamma \sim 10^{-5}$ yr $^{-1}$ per galaxy (Amaro-Seoane et al., 2023; Fragione et al., 2018). The volumetric rate is $\mathcal{R} \sim 0.1$ – 10 Gpc $^{-3}$ yr $^{-1}$.

Comparison with LIGO/Virgo O4. Current LIGO/Virgo/KAGRA O4 results constrain the IMBH merger rate to $\mathcal{R} \lesssim 0.1$ – 1 Gpc $^{-3}$ yr $^{-1}$ for $m_1 + m_2 \sim 100$ – $1000 M_\odot$ (LVK Collaboration, 2024). Our prediction of $\mathcal{R} \sim 0.1$ – 10 Gpc $^{-3}$ yr $^{-1}$ is in tension with the upper end of this range. This provides an immediate observational constraint: if O4 limits tighten to $\mathcal{R} < 0.1$ Gpc $^{-3}$ yr $^{-1}$, the DCBH population must be reduced below our fiducial estimate. We note that our conservative seed density ($n = 0.1$ Mpc $^{-3}$) predicts $\mathcal{R} \sim 0.3$ Gpc $^{-3}$ yr $^{-1}$, which is marginally compatible with current limits. Further O4 data will sharpen this test.

Microlensing (Roman). The *Nancy Grace Roman Space Telescope* (Spergel et al., 2015) will detect long-duration microlensing events ($t_E \sim 100$ – 1000 days; Wyrzykowski & Mandel 2020). The predicted optical depth toward M31 is $\tau_{\text{ML}} \sim 10^{-7}$ – 10^{-6} (Calcino et al., 2018).

Table 5: Detectability predictions for the baryonic dark sector.

Facility	Target	Observable	Predicted signal
ngVLA	Cold gas	Molecular absorption (CH, OH)	$\Delta v \sim 0.1$ km s $^{-1}$; 1 per 10–50 lines
SKA	Cold gas	HI 21-cm absorption	$\tau \sim 10^{-3}$ – 10^{-2}
SKA	Cold gas	FRB DM variance	$\delta\text{DM} \sim 1$ – 10 pc cm $^{-3}$
LISA	DCBHs	IMBH mergers	$\sim 10^{-5}$ yr $^{-1}$ per galaxy
<i>Roman</i>	All BHs	Microlensing toward M31	$\tau_{\text{ML}} \sim 10^{-7}$ – 10^{-6}
O4/O5	DCBHs	IMBH merger rate	$\mathcal{R} \sim 0.3$ Gpc $^{-3}$ yr $^{-1}$

Counterarguments and Responses

Objection: The IMBH merger rate at the upper end of the predicted range is already in tension with O4 limits.

Response: This is correct and constitutes a real-time falsification channel. Our conservative fiducial ($n = 0.1 \text{ Mpc}^{-3}$) predicts a rate near the current upper limit. If O4/O5 data exclude $\mathcal{R} > 0.1 \text{ Gpc}^{-3} \text{ yr}^{-1}$, the DCBH seed density must be further reduced—but as shown in §7, the model does not require DCBHs to close the mass budget.

Section Result and Implications

Every component has quantitative, falsifiable observational signatures. The LIGO/Virgo O4 limits already constrain the DCBH population and will sharpen with O5.

11 Discussion

11.1 Relationship to the missing-baryon problem

The baryonic dark sector ($\sim 4.6 \times 10^{10} M_{\odot}$ per $L_{\star}$ galaxy) closes the gap between the observed stellar mass and the expected baryonic content of individual halos (Bregman, 2007; Shull et al., 2012). It does not address the intergalactic missing-baryon problem.

11.2 Comparison with MACHO constraints

The EROS-2 (Tisserand et al., 2007) and MACHO (Alcock et al., 2000) surveys excluded compact objects in the range 10^{-7} – $10 M_{\odot}$ as the dominant halo dark matter. The DCBH population (~ 100 – $5000 M_{\odot}$) falls above this range. Stellar-mass black holes (~ 10 – $80 M_{\odot}$) are at the upper edge, where constraints are weak. Brown dwarfs ($\sim 0.04 M_{\odot}$) are within the excluded range but constitute only $\sim 13\%$ of the remnant budget ($\sim 9\%$ of the total baryonic dark sector).

11.3 Implications for the vacuum capture model

The three-component decomposition—vacuum (dominant, ~ 80 – 90%), remnants (~ 7 – 13%), cold gas (~ 3 – 5%)—is internally consistent. The exact partition depends on σ_v : at $\sigma_v = 170 \text{ km s}^{-1}$, the vacuum accounts for $> 90\%$ and baryonic corrections are minor; at $\sigma_v = 150 \text{ km s}^{-1}$, baryons contribute $\sim 20\%$.

11.4 Comparison with CDM predictions

In Λ CDM, the dark halo is entirely non-baryonic. The vacuum capture model inverts this: the dominant dark component is vacuum energy, and the baryonic dark sector plays the role of CDM “sub-structure.” The key difference is that the baryonic dark sector is composed of *known* objects rather than hypothetical particles.

12 Limitations

1. The DCBH population remains the most uncertain component. Even with the conservative fiducial ($n = 0.1 \text{ Mpc}^{-3}$), the range spans $0.15\text{--}2.0 \times 10^{10} M_{\odot}$.
2. The fractal cold-gas model relies on a turbulence-driving argument that is order-of-magnitude. A self-consistent magnetohydrodynamic simulation of fractal halo gas has not been performed. The survival of individual cloudlets against thermal conduction in tangled (not ordered) magnetic fields is an open question.
3. The radial distribution of remnants and gas is assumed, not derived self-consistently from the galactic potential. The shape mismatch between the available and required profiles (Table 4) may indicate that the remnant or gas profiles need refinement.
4. Galaxy-by-galaxy rotation-curve fits have not been performed.
5. The vacuum baseline uses the isothermal ansatz, which may break down in the outer halo (beyond $\sim 100 \text{ kpc}$). The 200 kpc shortfall (Table 4) may reflect this limitation rather than a deficit in the baryonic sector.
6. Mass lost to intergalactic space through galactic winds and outflows is not accounted for.
7. The total stellar mass formed depends on the assumed IMF at all epochs. The fiducial intermediate slope ($\alpha_3 = -1.5$) is a compromise; the true value is unknown.

13 Future Directions

1. **Galaxy-by-galaxy rotation-curve fits.** Apply the three-component model to the SPARC (Lelli et al., 2016) and LITTLE THINGS (Hunter et al., 2012) samples.
2. **MHD simulations of fractal halo gas.** Test whether cosmological accretion and galactic fountain flows can sustain a fractal hierarchy at 15–80 kpc.
3. **LIGO/Virgo O5 constraints.** Compare updated IMBH merger rate limits against the DCBH population model.
4. **ngVLA absorption-line survey design.** Specify the optimal number of background sources and impact parameters.
5. **Sensitivity of the vacuum profile at large radii.** Develop a non-isothermal vacuum profile for $r > 100 \text{ kpc}$ to address the 200 kpc shortfall.
6. **Cross-correlation with CGM observations.** Compare with COS-Halos (Werk et al., 2014) and CGM² (Wilde et al., 2021).
7. **Distinguish DCBHs from primordial black holes.** Through spatial distribution and mass function differences.

14 Position Within the Research Programme

This article is the combined Papers #14 and #15 in the programme *What If the Vacuum Gravitates Locally?* It completes the galactic mass budget initiated in Paper #12.

The original programme separated the two components into Paper #14 (cold gas) and Paper #15 (compact remnants). We merged them because: (i) the constraint is

on the *total* baryonic dark mass; (ii) remnants and gas are anti-correlated radially; (iii) detection strategies are best presented together.

All completed preprints are available at:

<https://interdisciplinary-research.institute/cosmology-and-theoretical-physics/>

15 Recommended Journal for Publication

This article is recommended for submission to **The Astrophysical Journal**.

The paper constructs a quantitative census of the baryonic dark sector in galactic halos and demonstrates consistency with the vacuum capture model's rotation-curve decomposition. The content bridges stellar evolution, ISM physics, gravitational-wave astrophysics, and galactic dynamics. *The Astrophysical Journal* regularly publishes work on stellar remnant populations (Adams & Laughlin, 2006; Chabrier, 2003), circumgalactic gas (Miller & Bregman, 2015; Werk et al., 2014), compact-object demographics (Arca Sedda et al., 2021), and dark-matter alternatives (McGaugh et al., 2016).

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A Numerical Integration of the Remnant Census

The following Python code computes the remnant mass integral (Eq. 10) for the fiducial model. The IMF is properly normalized at each metallicity bin so that $\int_{0.01}^{m_{\max}} m \xi(m) dm = 1 M_{\odot}$.

```
import numpy as np
from scipy.integrate import quad

t0 = 13.0          # Gyr
Mstar_tot = 6e10 # Msun
Zsun = 0.014
tau_sfr, beta = 5.0, 0.5

norm_sfr = quad(
    lambda t: (t/t0)**beta * np.exp(-t/tau_sfr),
    0, t0)[0]
A_sfr = Mstar_tot / norm_sfr

def sfr(t):
    return A_sfr * (t/t0)**beta * np.exp(-t/tau_sfr)

def Z_of(t):
    age = t0 - t
    if age < 0.5: return Zsun * 1e-3
    feh = min(0.3, -1.0 + 0.8*np.log10(age))
    return Zsun * 10**feh

def alpha3(Z):
    if Z < 1e-4*Zsun: return -1.5
    if Z < 1e-2*Zsun: return -1.8
    return -2.3

def mmax(Z):
    return 300.0 if Z < 1e-4*Zsun else 150.0

def tau_ms(m):
    return 10.0 * m**(-2.5)

def m_rem(mi, Z):
    if mi < 0.03: return mi * 0.65 # turnover
    if mi < 0.08: return mi
    if mi < 0.9:  return 0.0
    if mi < 8.0:  return 0.08*mi + 0.489
    if mi < 25.0: return 1.35
    if Z < 0.01*Zsun:
        return max(5.0, 0.9*mi - 5.0)
    return max(5.0, 0.4*mi - 3.0)

def imf_xi(m, a3):
    """Kroupa IMF with variable high-mass slope."""
    c1 = 0.08**(1.3-0.3)
    c2 = 0.50**(a3+1.3)
```

```

if m < 0.08:    return m**(-0.3)
elif m < 0.50: return c1 * m**(-1.3)
else:          return c1 * c2 * m**(a3)

def imf_norm(a3, mm):
    """Normalize so int(m*xi*dm) = 1 Msun."""
    mg = np.logspace(-2, np.log10(mm), 800)
    integrand = np.array(
        [m * imf_xi(m, a3) for m in mg])
    return np.trapz(integrand, mg)

dt = 0.1
times = np.arange(dt/2, t0, dt)
totals = {'WD':0, 'NS':0, 'BH_soZ':0,
          'BH_loZ':0, 'BD':0}

for t in times:
    Z = Z_of(t)
    a3 = alpha3(Z)
    mm = mmax(Z)
    norm = imf_norm(a3, mm)
    sfr_dt = sfr(t) * dt * 1e9

    mg = np.logspace(-2, np.log10(mm), 800)
    for i in range(len(mg)-1):
        m = 0.5*(mg[i]+mg[i+1])
        dm = mg[i+1]-mg[i]
        xi = imf_xi(m, a3) / norm
        if m >= 0.08 and tau_ms(m) > t:
            continue
        mr = m_rem(m, Z)
        if mr <= 0: continue
        w = sfr_dt * xi * dm * mr
        if m < 0.08:      totals['BD'] += w
        elif m < 8:      totals['WD'] += w
        elif m < 25:
            totals['NS'] += w * 0.80 # kick loss
        elif Z < 0.01*Zsun: totals['BH_loZ'] += w
        else:            totals['BH_soZ'] += w

for k,v in totals.items():
    print(f"{k}: {v:.2e} Msun")
print(f"Total: {sum(totals.values()):.2e} Msun")

```

Peer Reviews and Revision History

First Round

Reviewer recommendation: Major revision.

Reviewer assessment. The reviewer found the paper clearly written and well-structured, praising the “Logical Structure of the Argument” section and the systematic counterargument-and-response format. The detectability predictions were called “concrete and falsifiable.” However, the reviewer identified five major issues and four moderate issues.

Major Concern 1: The DCBH contribution is doing most of the heavy lifting, yet is the least constrained. The reviewer noted that the fiducial seed density $n_{\text{DCBH}} = 0.3 \text{ Mpc}^{-3}$ was at the upper end of the theoretical range and that the factor-of-two accretion growth lacked a physical model.

Response. We agree. The fiducial seed density has been reduced to $n_{\text{DCBH}} = 0.1 \text{ Mpc}^{-3}$, and the accretion growth factor has been removed from the fiducial estimate. The Bondi accretion rate for isolated $500 M_{\odot}$ black holes in the halo is now computed explicitly (§7): $\dot{M}_{\text{Bondi}} \sim 10^{-12} M_{\odot} \text{ yr}^{-1}$, giving negligible growth over 13 Gyr. The fiducial DCBH mass has decreased from $2.0 \times 10^{10} M_{\odot}$ to $0.65 \times 10^{10} M_{\odot}$. The model remains viable because (i) the σ_v sensitivity analysis (Table 1) shows the required residual can be as low as $\sim 2 \times 10^{10} M_{\odot}$, and (ii) DCBHs are no longer load-bearing: even without them, the baryonic budget ($\sim 4.0 \times 10^{10} M_{\odot}$) falls within the σ_v range.

Major Concern 2: The top-heavy IMF at low metallicity is treated as near-certain. The reviewer recommended centering the fiducial model on an intermediate slope rather than the most top-heavy option.

Response. Adopted. The fiducial high-mass slope has been changed from $\alpha_3 = -1.0$ to $\alpha_3 = -1.5$ at $Z < 10^{-4} Z_{\odot}$. The remnant yield enhancement is now $\sim 2\times$ (fiducial) rather than $\sim 3\times$ (previous). The $\alpha_3 = -1.0$ case is presented as an optimistic bound, and $\alpha_3 = -2.3$ as the conservative floor. All tables and summary numbers have been updated accordingly.

Major Concern 3: The fractal cold-gas model lacks a self-consistent survival argument. The reviewer asked what drives supersonic turbulence at 15–80 kpc in the halo.

Response. A new subsection (§8.5) has been added, identifying three mechanisms: cosmological accretion ($\dot{E} \sim 10^{41} \text{ erg s}^{-1}$), galactic fountain flows ($\dot{E} \sim 10^{40} - 10^{41} \text{ erg s}^{-1}$), and ram pressure/satellite interactions ($\dot{E} \sim 10^{39} - 10^{40} \text{ erg s}^{-1}$). The total available power exceeds the estimated dissipation rate by a factor of $\sim 3-10$. We now explicitly state that this is an order-of-magnitude argument and that the steady-state assumption remains to be validated by MHD simulations. The reviewer’s point about tangled vs. ordered magnetic fields is acknowledged: the conduction suppression factor is reduced in tangled fields, and cloudlet survival is flagged as an open question.

Major Concern 4: The “target” residual is defined by an unpublished model. Sensitivity to σ_v is needed. *Response.* A sensitivity analysis has been added (Table 1 and §3.3). The required residual varies from $\sim 2 \times 10^{10} M_\odot$ at $\sigma_v = 170 \text{ km s}^{-1}$ to $\sim 9 \times 10^{10} M_\odot$ at $\sigma_v = 150 \text{ km s}^{-1}$. The baryonic budget ($4.6 \times 10^{10} M_\odot$ with range $2.2\text{--}8.8 \times 10^{10} M_\odot$) overlaps this band at all observationally allowed values.

Major Concern 5: The model doesn’t fit well at 200 kpc. *Response.* The 200 kpc shortfall is now addressed as a substantive issue in a dedicated subsection (§9.3). Two possible resolutions are discussed: (i) the vacuum profile may be steeper than isothermal beyond $\sim 100 \text{ kpc}$; (ii) the cold gas may extend beyond 80 kpc. We no longer dismiss the discrepancy as “within error bars.”

Moderate Concern 6: Tension with LIGO/Virgo O4. *Response.* A comparison with current LIGO/Virgo/KAGRA O4 upper limits has been added to §10. Our conservative fiducial predicts $\mathcal{R} \sim 0.3 \text{ Gpc}^{-3} \text{ yr}^{-1}$, which is marginally compatible with current limits. The IMBH merger rate is identified as a real-time falsification channel.

Moderate Concern 7: Neutron star retention correction. *Response.* Clarified. The NS population of $\sim 1.2 \times 10^9$ and mass of $0.17 \times 10^{10} M_\odot$ in Table 3 are post-kick-correction values. This is now stated explicitly in the text.

Moderate Concern 8: Brown dwarf overestimate. *Response.* The sub-stellar mass function turnover below $\sim 0.03 M_\odot$ (Kirkpatrick et al., 2021) has been incorporated. Brown dwarf mass reduced from 0.60 to $0.40 \times 10^{10} M_\odot$.

Moderate Concern 9: Cosmic baryon fraction argument is circular. *Response.* Accepted. The section now explicitly presents the baryon-fraction agreement as a “consistency check,” not as an independent prediction.

Minor concerns: (i) SFR convention clarified: t is look-back time. (ii) Intermediate steps for the gas integral are now shown explicitly. (iii) Rounding convention stated in Table 3 caption. (iv) Companion paper DOIs confirmed accessible. (v) Appendix A code now includes proper IMF normalization and is fully executable.

Summary of changes in revision.

1. Reduced DCBH fiducial to $n = 0.1 \text{ Mpc}^{-3}$; removed accretion growth; added explicit Bondi rate calculation.
2. Changed fiducial IMF slope to $\alpha_3 = -1.5$; $\alpha_3 = -1.0$ presented as optimistic bound.
3. Added turbulence-driving subsection with energy budget estimates from three mechanisms.
4. Added σ_v sensitivity analysis (Table 1).
5. Addressed 200 kpc shortfall as substantive issue with two possible resolutions.
6. Added LIGO/Virgo O4 comparison for IMBH merger rates.

7. Clarified NS retention correction (post-kick values throughout).
8. Reduced brown dwarf mass following Kirkpatrick et al. (2021) turnover.
9. Reformulated cosmic baryon fraction as consistency check.
10. Fixed SFR time convention, added intermediate gas-integral steps, stated rounding convention.
11. Rewrote Appendix A code with proper IMF normalization.
12. All fiducial numbers, tables, and abstract updated for revised parameters.